



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Final Exam Summer 2025

CSE 2233/CSI 233: Theory of Computation/Theory of Computing

Total Marks: 40

Duration: 2 Hours

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. (a) Consider the following Context-free Grammar over the alphabet $\{0, 1, 2\}$:

$S \rightarrow ABCD$
 $A \rightarrow 00A1 \mid \epsilon$
 $B \rightarrow 1B0 \mid 10$
 $C \rightarrow 0C2 \mid 22$
 $D \rightarrow 2D \mid 0$

- i. Show a leftmost derivation of the string **001110000222220** on the grammar. [2]
ii. Show a rightmost derivation of the string **00000011110220** on the grammar. [2]

- (b) Consider the following Context-free Grammar over the alphabet $\{a, b, c\}$:

$S \rightarrow ABC \mid DE$
 $A \rightarrow aA \mid \epsilon$
 $B \rightarrow aBb \mid \epsilon$
 $C \rightarrow bCc \mid Cc \mid \epsilon$
 $D \rightarrow aA \mid a$
 $E \rightarrow bEc \mid \epsilon$

Show that the grammar is ambiguous, by showing two parse trees of the string **aaabbbbccc** on the grammar. [4]

2. Design CFGs that generate the following languages: [4x2]

a) $L = \{ w \in \{a,b\}^* \mid a^i b^j \text{ where } i = 2j + 2 \text{ and } i, j > 0 \}$

b) $L = \{ w \in \{a,b\}^* \mid \text{the 2nd and 2nd-last characters are the same} \}$

c) $L = \{ w \in \{a,b\}^* \mid (x^n y^{2n})^{2m} z^m \text{ where } m, n > 0 \}$

d) $L = \{ w \in \{a,b\}^* \mid w \text{ is a palindrome of even length} \}$

3. Convert the following grammars into Chomsky Normal Form (CNF):

[4x2]

a) $E \rightarrow E - T$

$$T \rightarrow R \mid T / F$$

$$R \rightarrow F \mid T - E \mid \epsilon$$

$$F \rightarrow a \mid b \mid Fa \mid Fb$$

b) $S \rightarrow ABA$

$$A \rightarrow bA \mid AAB \mid \epsilon$$

$$B \rightarrow BAB \mid aB$$

4. Draw the Push Down Automata (PDA) for the following languages:

[4x2]

a) $L = \{ a^x b^y \# c^z \mid \Sigma = \{ a, b, c, \# \} ; y = x/2 \text{ or } z = 2y + 1 ; x, y, z \geq 0 \}$

b) $L = \{ w^x : = v^y \mid \Sigma = \{ w, v, :, = \} ; y \geq 3x ; x, y \geq 1 \}$

5. a) Draw a Turing Machine for the following language

[4x2]

$$L = \{ a^p b^{2q} c^r \mid \Sigma = \{ a, b, c \} ; r - q = 2p ; p, q, r \geq 1 \}$$

b) Consider the following Turing Machine (Here B means Blank Symbol). With the help of Tape Traversal, validate the input: **BabaabbbbaB**

